

Part A.

Application: Product Recommendation

Question	Answer
Problem	Shorten the time in product researching through product suggestion and recommendation algorithms.
Inputs	Purchase history, liked and added-to-the-cart products/item, search history
Outputs	Suggested items will be displayed on online shopping applications or videos about product review and recommendation will come across the user's algorithm.
Pattern	Users that purchase items, or just even add-to-cart an item, will have similar product recommendations based on category. For example school supplies. If a person purchased a yellow pad online, the user's online shopping app will be displaying ballpens, bags, and any other school supplies in the product recommendation section.
Data	Users liked products, activity and interaction in the item/product.

Part B. Traditional Programming vs Machine Learning

Traditional Programming

- Product recommendation algorithm will be extremely difficult when it is approached with traditional programming in which will be through: Input -> Algorithm -> Output. In order for this to work, the programmer will have to write the rules or instructions using IF/THEN. For example, if the user purchased a notebook, then recommend a ballpen. But if there were a lot of school supplies, it would take some time to finish the rules or instructions.

Machine Learning

- In order for the ML algorithm to work in this product recommendation, it should learn from data such as purchase history. For example, we have a data of purchase history that contains notebook, ballpen, pencil — this purchase history can be referred to the transaction made by Customer 1. Then, Customer 2 has an order of notebook, ballpen, eraser, pencil. Through training and testing, the ML algorithm will learn that notebooks often come with a ballpen, hence, will recommend a ballpen once a notebook was purchased by another customer.

Part C. Create Your Own Dataset

Since the application I chose was Product Recommendation, and the example was based on school supplies purchases, the dataset should be:

Input = number of notebooks purchased

Output = number of pens recommended

Input	Output
1	2
2	4
3	6
4	8
5	??

1. The Pattern

- Whenever a notebook is purchased, it is recommended with two (2) pens — it might be the user's discretion to make sure an extra pen will be available for emergencies.

2. Determine the missing output.

- Since the dataset shows that for every notebook, there are recommended 2 pens, the missing output in the fifth row should be ten (10) pens.

3. Write the mathematical function.

$Y = 2x$. Y is the output and X is the input.

4. Explain how a machine learning algorithm could learn this relationship from the examples.

- A machine learning algorithm could learn the relationship between the input and output by training and testing. Through this process, the machine learning algorithm can compare that the input consistently provides twice the output by using the mathematical function $y=2x$ — the number of pen recommendations.

Part D. Explain the Training Process

Machine Learning Algorithm learns from data by repeatedly examining the pattern given in a dataset and through the process of training and testing. In this example that we have about product recommendation, the user input the data by adding-to-cart the item they are interested in or simply by checking it out/purchasing. The ML compares what other item was checked-out (ex. ballpen) — which shows the output. Then, after a lot of transactions, the ML may discover that users who buy notebooks usually buy ballpens as well — learning the relationship. So when a new user buys a notebook, the application can predict what product should be displayed on the recommendation section..

Reflection

Learning from data is more useful because what you examine is patterns generated by users across the globe. It is progressive because while you are training and testing the data, the machine learns from it at the same time. It also saves much time, cost, and effort, especially to big technological industries because you're training the machine to think and predict the next move. In simple terms, learning from data is proactive since we are preparing for what is next, while manually programming every possible rule is reactive — reactive to what happened. Just like the example in this activity, the machine will predict what item to display through the I/O of the user. It actually saves much time and effort rather than writing an IF/THEN program.